

VECTOROPS

Technical Disclosure v1.0

Mission-aware adaptive post-processing of sensor video for search-and-rescue operations

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Document date: August 10, 2026

Public technical disclosure. This document describes implemented reference embodiments and contemplated alternatives. It does not include production source code, credentials, client information, or mission data.

Contents

1. Abstract	3
2. Authors, Conception, and Disclosure History	3
3. Technical Field and Operational Problem	3
4. System Architecture	4
4.1 Reference deployment arrangements	4
5. Mission and Evidence Data Model	4
6. Adaptive Deadline-Based Discovery Processing	5
6.1 Runtime measurement and projection	5
6.2 Progressive workload reduction	5
6.3 Stability and recall safeguards	5
6.4 Other adjustable resources	6
7. Rule-Based Discovery and Candidate Scheduling	6
7.1 Tiered scoring	6
7.2 Spatially fair computational budgeting	6
8. Multi-Stage Event Refinement	6
8.1 Duplicate grouping	7
8.2 Quilt or batched contextual review	7
8.3 Detailed event review	7
8.4 Focused selection	7
9. Operator-Directed Investigation	7
10. Confidence-Appropriate Localization and Display	7
11. Telemetry, Geographic Data, and Operational Export	8
11.1 TAK and SARTopo embodiments	8
12. Streaming, Hosted, and Cloud-App Extensions	8
13. Example Methods	9
13.1 Deadline-adaptive mission method	9
13.2 New-information investigation method	9
14. Contemplated Variations	9
15. Numbered Disclosure Aspects	9
16. Rights, Integrity, and Scope of Publication	10
Appendix A. Release Checklist	11

1. Abstract

VectorOPS is a mission-aware system for post-processing recorded or streaming sensor video into operator-reviewable detection events and operational findings. A reference embodiment uses a rule-based computer-vision discovery pass, duplicate grouping, batched contextual artificial-intelligence review, detailed event review, focused-event selection, and optional operator-directed investigation. The system preserves relationships among a source frame, detection geometry, contextual imagery, analytical results, timestamps, telemetry, geographic information, and lifecycle state.

A principal mechanism dynamically adjusts discovery workload so processing can complete within an operational interval, such as the duration of a drone flight or the expected arrival time of the next recording. The mechanism measures processing performance, projects completion time, and progressively changes spatial resolution, temporal sampling, candidate budgets, or analysis depth while applying safeguards that preserve useful recall. The resulting findings can be reviewed locally and distributed to operational systems including TAK and SARTopo.

Core concept: retain mission evidence, adapt computational effort to an operational deadline, refine detections through staged analysis, and present each finding at a precision justified by its evidence.

2. Authors, Conception, and Disclosure History

David Boyce and Laetitia Gerard jointly developed the disclosed VectorOPS concepts. The earliest presently identified conception or implementation date is May 2, 2026. This document records the implemented reference embodiment reviewed in August 2026 together with reasonable implementation alternatives contemplated by the authors.

Date	Disclosure event	Access or audience
May 14, 2026	VectorOPS functionality video	Public YouTube publication
June 9, 2026	VectorOPS functionality video	Public YouTube publication
June 13, 2026	Laptop demonstration	Pete and Darian, Converse Police Department; no installed copy supplied
July 28, 2026	VectorOPS functionality video	Public YouTube publication
August 10, 2026	Technical Disclosure v1.0	Prepared for public GitHub Release

3. Technical Field and Operational Problem

The disclosure relates to computer-implemented processing of video, imagery, sensor data, and associated metadata. Example fields include search and rescue, disaster response, public safety, wildfire response, inspection, security, and other operations in which sensor recordings must be converted into prioritized, geographically meaningful findings.

- A drone may produce a long recording during each battery-limited flight, while the next flight creates another recording before operators can manually review the first.
- Fixed-rate processing may either finish too slowly or sacrifice unnecessary image detail when more resources are available.
- First-pass detections may be numerous, duplicated across nearby samples, or based on visually complex terrain.
- A clue may lie near, rather than within, an initial detection region; later contextual analysis may classify a useful finding without reliably localizing it.
- New witness reports or mission details may arrive after the original video has already been processed.
- Operational users need concise evidence and geographic context in tools they already use rather than a replacement incident-management platform.

4. System Architecture

A reference system includes a native sensor-processing layer, an application and event-lifecycle layer, one or more AI-analysis services, persistent mission storage, an operator interface, and operational export adapters. The components may execute on one workstation or be distributed across edge and hosted resources.



Subsystem	Reference responsibilities
Sensor input	Read a recorded file, RTSP stream, camera device, extracted frames, embedded metadata, or sidecar telemetry.
Discovery processor	Sample frames; detect SAR colors, thermal structures, anomaly candidates, and person-sized regions; produce event evidence.
Event lifecycle	Group duplicates; persist frames, crops, context, timing, confidence, reasons, and stage transitions.
AI review	Perform batched contextual screening, detailed event analysis, and operator-directed investigation.
Operator interface	Search, sort, review, annotate, investigate, validate, and suppress misleading graphical precision.
Operational adapters	Package reviewed findings for TAK, SARTopo/CalTopo, GIS, communications, or cloud applications.

4.1 Reference deployment arrangements

- Local/edge: discovery and review occur on a field workstation without requiring persistent connectivity.
- Hybrid: discovery runs locally while selected image evidence is sent to a hosted AI service.
- Hosted: uploaded mission evidence is queued and processed by scalable services, with authenticated operator access.
- Streaming: RTSP or camera frames are segmented or buffered so the same event lifecycle can operate on a continuing source.

5. Mission and Evidence Data Model

The system organizes evidence by mission. A mission may contain source recordings, segments, extracted frames, raw frames, thumbnails, contextual images, analytical metadata, AI results, telemetry, geographic coordinates, and an event database. A source-derived identifier and stage history preserve provenance as later analysis refines the event.

Event information	Examples
Source identity	Mission, file or stream, segment, frame/sample number, time offset, capture time
Detection evidence	Original box or region, crop, full frame, contour/heat map, detector scores, target color
Contextual results	Person, fabric, anomaly, color, likely class, confidence, reasons, evidence source
Location evidence	Original coordinates, AI region, location confidence, telemetry position, map position
Lifecycle	Discovery, duplicate grouping, quilt review, detailed review, focused decision, investigation, export
Operator state	Reviewed, selected, confirmed, dismissed, notes, prompt, export status

Analytical metadata for a frame is treated as an atomic packet. A reader should not retry an undersized buffer in a way that advances to a subsequent frame and incorrectly associates later metadata with earlier imagery. Implementations may allocate a sufficiently large packet buffer, use length-prefixed messages, retain frame identifiers during retry, or use another atomic transport.

6. Adaptive Deadline-Based Discovery Processing

The reference embodiment attempts to finish discovery processing within the duration of the source video. This models an operational objective in which processing for one drone flight should finish before the next flight's recording becomes available. Other embodiments use remaining battery life, expected upload time, a communications window, a response deadline, or an operator-entered completion target.



6.1 Runtime measurement and projection

For the active processing level, the system accumulates elapsed processing time and the number of completed samples. It derives an average time per current-level sample. From source duration, segment size, temporal-sampling policy, and completed work, it estimates the number of remaining samples.

Projected completion = max(accumulated discovery time, elapsed wall-clock scan time) + average current-level sample time × remaining current-level samples.

Using the greater of accumulated discovery time and elapsed wall-clock time accounts for media access, setup, coordination, and similar costs that may not be captured by frame-analysis timing alone. The projection is compared with the selected operational interval.

6.2 Progressive workload reduction

Illustrative level	Spatial policy	Temporal policy for a 15-second segment
0-1	Configured or original discovery resolution	Up to one sample per second
2	Downscale discovery processing to a maximum height such as 720 pixels	Retain full temporal sampling
3	Reduced spatial processing	Three samples, for example at 0, 5, and 10 seconds
4	Reduced spatial processing	Two samples, for example at 0 and 10 seconds
5	Reduced spatial processing	One sample, for example at 0 seconds

When projected completion exceeds the deadline and unfinished discovery work remains, the system increases the reduction level. It then recalculates which samples remain required under the new policy. Previously completed samples are retained and samples no longer required are not processed again.

6.3 Stability and recall safeguards

- Require multiple measurements at a newly selected level before reducing again, preventing a single slow sample from causing cascading reduction.
- Reset level-specific timing when the active reduction level changes so measurements from a higher-cost level do not distort the next decision.
- Cap reduction for short prerecorded tests, which could otherwise lose excessive temporal recall before stable measurements are available.
- Delay adaptive decisions while a frame-cache extraction process is active when extraction overhead would inflate discovery timing.
- Do not reduce after relevant discovery work is complete, because later AI or operator work cannot be accelerated by changing frame sampling.

- Permit fixed resolution, fixed sample interval, or disabled adaptation when an administrator requires deterministic coverage.

6.4 Other adjustable resources

The feedback method is not limited to resolution and frame sampling. An implementation may adjust candidate count, region-of-interest sizes, number of detector classes, expensive-score quotas, AI-review depth, batch size, model selection, image compression, cloud concurrency, or combinations of these parameters.

7. Rule-Based Discovery and Candidate Scheduling

A reference color-discovery embodiment converts frames into color spaces and forms masks for configurable SAR-relevant colors. Morphological operations remove isolated noise and join nearby evidence. Candidate regions are evaluated at multiple person- or object-sized windows. Thermal and anomaly embodiments may provide additional candidate types.

7.1 Tiered scoring



- An ultra-cheap pass uses integral images or equivalent precomputed statistics to reject low-density windows without repeated per-window image scans.
- A cheap pass considers color density, saturation, shape, orientation, texture, vegetation penalties, or similar features.
- A bounded expensive pass evaluates only candidates surviving earlier stages.
- Non-maximum suppression and center-distance suppression reduce overlapping or near-duplicate candidate regions.
- Coordinates measured on a reduced processing image are scaled to the original source frame before evidence is persisted.

7.2 Spatially fair computational budgeting

Instead of selecting all expensive candidates from the globally strongest texture cluster, the system assigns candidates to coarse spatial buckets and limits expensive evaluations per bucket. A reserve may first admit high-value SAR-color clusters. A spatially fair pass then admits candidates across buckets, followed by a bounded fill pass. This preserves geographic coverage within the frame while keeping processing cost predictable.

A processing budget is allocated across image regions so that one high-texture terrain area cannot consume all expensive analysis capacity.

8. Multi-Stage Event Refinement

The reference event pipeline separates inexpensive discovery from increasingly selective contextual review. Each stage records status, time, decision, confidence, and reason so the final result can be traced to earlier evidence.



8.1 Duplicate grouping

Detections from nearby times, overlapping regions, related source frames, or visually similar evidence may be grouped before AI analysis. A representative event and its context are retained while redundant AI requests are avoided. Distinct raw frames may remain separately eligible when temporal separation or evidence differences are operationally meaningful.

8.2 Quilt or batched contextual review

Multiple candidate crops may be arranged into a labeled composite image or submitted as a batch. An AI service evaluates relative visual groups, likely classes, thermal or color priority, uniqueness, and whether an event deserves detailed review. The batch stage reduces cost and latency while permitting comparison among events from the same mission.

8.3 Detailed event review

Events promoted by the batch stage receive a detailed analysis using an individual crop, full-frame context, or both. Example categories include person, fabric, anomaly, equipment, and SAR-relevant colored objects. Each category may store a state, confidence, and explanation independently rather than collapsing all evidence into one score.

8.4 Focused selection

A lifecycle evaluator decides whether an event belongs in a focused operator queue. The decision may combine detector scores, batch results, detailed results, confidence thresholds, reason codes, mission configuration, and operator policy. Events excluded from the focused queue remain available in the mission record.

9. Operator-Directed Investigation

Mission details often change after a recording has been processed. VectorOPS permits an operator to provide newly reported facts and apply them to retained event evidence. Examples include clothing color, jacket or fabric description, equipment, subject characteristics, clue type, last-known-area information, or a free-form mission-specific description.

- The investigation may operate on a selected event, a focused set, or all retained mission evidence.
- The system can submit the original detection crop, full-frame context, temporal neighbors, or a composite of these sources.
- Results record whether the requested detail matched, confidence, reasoning, useful evidence sources, and an optional refined region.
- Newly supplied criteria can be evaluated without rerunning the original full-video discovery pass.
- Prompts may be generated from structured fields, operator narrative, mission templates, or combinations of these inputs.

Retained mission evidence becomes reusable: new search facts can be applied after discovery, while the original detector evidence and provenance remain intact.

10. Confidence-Appropriate Localization and Display

Classification confidence and localization confidence are treated as separate properties. A later analysis may confidently identify a person or clue in the full-frame context while returning a region that does not enclose the identified object. Displaying that region as a precise box could mislead an operator.

Evidence condition	Illustrative display behavior
Reliable original detection geometry	Display the native detection region in discovery and ordinary playback views.
Valid refined region with sufficient location confidence	Position the finding using the refined region; optionally display geometry in review contexts.

Evidence condition	Illustrative display behavior
Positive classification but unreliable refined geometry	Suppress boxes and arrows; display concise text near the refined or original location.
No valid refined region	Anchor text at the original detection location and retain the contextual explanation.
Negative or terrain-only result	Do not draw a positive finding box; retain audit information in event details.

Reliability policies may consider coordinate validity, location-confidence thresholds, region dimensions, overlap or distance from the original detection, agreement among models, object presence within the returned region, crop-versus-context contribution, and operator confirmation. A conservative embodiment suppresses Investigation geometry in playback while preserving positive text.

11. Telemetry, Geographic Data, and Operational Export

Telemetry and geographic data may be read from metadata embedded in the video, a separate synchronized file, a flight log, a live sensor interface, a network message, or operator input. A synchronization function associates a frame time with an exact or interpolated position, altitude, orientation, camera pose, and related sensor state when available.

- When complete camera geometry is available, an image-space region may be projected to a ground estimate or uncertainty area.
- When only platform coordinates are available, the platform position and uncertainty are retained without representing an unsupported object location.
- When no location exists, the event remains useful as time- and frame-linked evidence without inserting placeholder location text into the visual finding.
- Exports may include event identifiers, time, geographic point or area, confidence, concise label, thumbnail, full-frame evidence link, notes, and provenance.

11.1 TAK and SARTopo embodiments

An export adapter transforms a reviewed VectorOPS event into the target system's supported object or message representation. TAK embodiments may use Cursor-on-Target or another supported interface. SARTopo/CalTopo embodiments may create markers, objects, tracks, or linked evidence using supported APIs or file formats. The adapter records delivery state and preserves the VectorOPS event identifier for later reconciliation.

The disclosed contribution is not the existence of a TAK or SARTopo connection by itself. It is the evidence-preserving transformation of staged sensor findings into confidence-appropriate operational objects without requiring the receiving team to adopt VectorOPS as its incident-management system.

12. Streaming, Hosted, and Cloud-App Extensions

- RTSP input can be decoded through a managed media process, auto-probed for resolution, segmented into rolling mission-aware storage, and submitted to the same queue used for files.
- A hosted deployment can allocate discovery or AI workers according to mission priority, deadline, source duration, and available compute resources.
- A cloud application can expose flagged events through authenticated organizational identities, including Microsoft 365 or equivalent identity providers.
- A Power Apps or other low-code client can display approved event summaries while source evidence and processing services remain in controlled storage.
- Offline operation can queue exports and synchronize them when connectivity returns.

13. Example Methods

13.1 Deadline-adaptive mission method

- Receive a sensor recording and determine an operational completion interval.
- Process samples under an initial spatial and temporal policy.
- Measure current-level processing performance and estimate remaining work.
- Project total completion time using elapsed wall time and estimated remaining work.
- When the projection exceeds the interval, select a lower-cost policy subject to stability and recall safeguards.
- Recompute unfinished work without discarding completed evidence.
- Persist detection events and route selected events to staged contextual review.

13.2 New-information investigation method

- Persist event crops, contextual frames, source identity, timing, and earlier analytical results.
- Receive a new mission-specific description from an operator.
- Select retained evidence relevant to the description.
- Analyze the evidence for the newly described subject, object, color, or clue.
- Store classification confidence separately from location confidence.
- Display a concise finding at a justified location while suppressing unsupported geometry.
- Optionally distribute the reviewed finding to an operational mapping system.

14. Contemplated Variations

Dimension	Illustrative alternatives
Platform	Drone, aircraft, vehicle, body-worn camera, fixed camera, robot, satellite-derived imagery
Sensor	RGB, thermal, infrared, multispectral, depth, radar-derived imagery, fused sensors
Processing	Real-time, near-real-time, post-flight, scheduled batch, operator-triggered
Compute	Laptop, edge appliance, vehicle server, cloud worker, distributed hybrid
Deadline	Video duration, battery interval, next upload, response SLA, bandwidth window, operator target
Analysis	Rules, classical computer vision, one AI model, model ensemble, human review, combinations
Finding display	Text, point, box, polygon, heat map, uncertainty area, ranked list, evidence-only record
Destination	TAK, SARTopo/CalTopo, GIS, CAD, messaging, Power Apps, evidence management

15. Numbered Disclosure Aspects

1. A computer-implemented method that adjusts analysis workload by comparing projected sensor-video processing completion with an operational interval.
2. The method of aspect 1 in which workload is adjusted by changing spatial resolution, temporal sampling, candidate budgets, analysis depth, or a combination thereof.
3. The method of aspect 1 in which the operational interval is based on source duration, remaining flight time, expected next upload, communications availability, or an operator deadline.

4. The method of aspect 1 in which timing is reset after a workload-level change and multiple samples are required before a subsequent change.
5. A candidate scheduler that distributes a bounded number of expensive image evaluations across spatial regions to preserve spatial diversity.
6. A staged event-analysis system comprising rule-based discovery, duplicate grouping, batched contextual review, detailed event review, and focused-event selection.
7. The system of aspect 6 further comprising operator-directed investigation using mission details received after initial processing.
8. The system of aspect 7 in which retained event crops and contextual frames are re-evaluated without repeating the full source-video discovery pass.
9. A finding-display method that stores classification confidence separately from localization confidence and suppresses graphical geometry when location precision is unsupported.
10. The method of aspect 9 in which finding text is positioned using a refined analysis region when reliable and an original detection location otherwise.
11. A mission evidence record that atomically associates a source frame with detection data, contextual evidence, lifecycle state, AI results, telemetry, and geographic information.
12. A system that transforms a reviewed mission event into an operational mapping object while retaining event provenance, confidence, time, and evidence references.
13. The system of aspect 12 in which the mapping destination comprises TAK, SARTopo, CalTopo, GIS, or another incident platform.
14. A mission-aware streaming embodiment that segments or buffers a continuing source and processes the resulting evidence through the same event lifecycle used for recorded files.
15. A hosted or hybrid embodiment that allocates processing according to mission priority, deadline, source duration, and available resources.

16. Rights, Integrity, and Scope of Publication

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This disclosure intentionally excludes production source code, executable credentials, API keys, private mission imagery, client information, and deployment secrets. The technical mechanisms are described through architecture, algorithms, data relationships, examples, and alternatives. Later releases may add newly developed subject matter and should be treated as supplements rather than retroactive changes to this version.

VectorOPS and associated branding, interfaces, and materials may be subject to additional intellectual-property rights. This technical report documents development history and is not legal advice or a representation that publication will have a particular legal effect in any proceeding.

Appendix A. Release Checklist

- Publish this fixed PDF as an asset of GitHub Release technical-disclosure-v1.0.
- Publish SHA256SUMS.txt with the release and verify the hash after upload.
- Enable immutable releases before or immediately after preparing the draft release, following GitHub's current workflow.
- Update the README disclosure table with the actual release date and release URL.
- Link the numbered release from relevant YouTube video descriptions.
- Preserve an offline copy of the release page, PDF, checksum file, and repository state.
- Publish later technical mechanisms as new numbered releases; do not replace v1.0 assets.